Coalescent Event Density across Sample Groupings Inferred from a Time Calibrated Phylogeny

Lintao Luo [1,*]

[1]giantlinlinlin@gmail.com

1 Materials and Methods

1.1 Node ages and equal sample size rarefaction

Analyses were carried out on a previously estimated time calibrated phylogeny of 8684 individuals, which was parsed and converted to node ages with DendroPy (v5.0.8) (Sukumaran and Holder 2010). Individuals were assigned to categories under five independent grouping schemes drawn from the sample metadata, and categories represented by fewer than 50 individuals were excluded. For a given category, coalescent events were defined as the internal nodes of the subtree induced by the sampled individuals of that category. An internal node was assigned to the induced subtree when at least two of its child subtrees each contained at least one individual of that category, a criterion that yields exactly $n-1$ nodes for a category of n individuals and that was asserted programmatically for every category. Because the shape of an induced subtree depends on the number of sampled individuals, all categories within a grouping scheme were randomly downsampled without replacement to a common size equal to 70 percent of the smallest category, and the procedure was repeated over 200 replicates under a fixed random seed (20260828). The downsampling target was deliberately set below the size of the smallest category so that every category, including the smallest one, was genuinely resampled. Had the target equalled the smallest category, that category would have been reported without the smoothing and without the resampling variance conferred on all other categories, which biases the comparison in its favour. Array operations were implemented with NumPy (v2.2.6) (Harris et al. 2020) and tabular processing with pandas (v2.3.3) (McKinney 2010) under Python (v3.10.19).

1.2 Adaptive kernel density estimation and standardisation

Node ages were converted to densities with an adaptive Gaussian kernel whose bandwidth at age a was taken as the larger of αa and a floor of 0.4 thousand years, where α was obtained by applying the Silverman rule (Silverman 1986) to the base 10 logarithm of all node ages of a grouping scheme and rescaling by the natural logarithm of 10. The floor declares the finest temporal scale that the analysis claims to resolve and suppresses the spurious peak that arises near the present, where the bandwidth is set by the floor rather than by the data. Each kernel was reflected at age zero so that its integral over the non negative half line equals unity, and densities were evaluated on a grid of 700 points per time window using probability density functions from SciPy (v1.15.2) (Virtanen et al. 2020). Densities were standardised to coalescent events per 1000 sampled individuals per thousand years, so that the area under each curve equals the number of events per 1000 individuals, an identity that was asserted for every replicate. Because the pointwise median across replicates does not commute with integration, the median curve and its accompanying interval were rescaled by a single scalar that restores the median of the replicate specific standardisation, a correction that remained below 4 percent throughout. Curves are reported as the pointwise median across replicates, and the shaded band spans the 5th to the 95th percentile of the replicates, so that it reflects resampling variation on a single fixed tree and does not propagate the uncertainty of the divergence time estimates. Ridgeline plots were generated with GGRIDGES (v0.5.7) (Wilke 2025) and GGPLOT2 (v4.0.3) (Wickham 2016), composed with PATCHWORK (v1.3.2) (Pedersen 2025) under R (v4.6.0) (R Core Team 2026), for two time windows spanning 0 to 10 and 0 to 60 thousand years.

References

Harris, C. R., K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, R. Kern, M. Picus, S. Hoyer, M. H. van Kerkwijk, M. Brett, A. Haldane, J. F. del Río, M. Wiebe, P. Peterson, P. Gérard-Marchant, K. Sheppard, T. Reddy, W. Weckesser, H. Abbasi, C. Gohlke, and T. E. Oliphant (2020). "Array Programming with NumPy". In: *Nature* 585.7825, pp. 357–362. DOI: 10.1038/s41586-020-2649-2.

McKinney, W. (2010). "Data Structures for Statistical Computing in Python". In: *Proceedings of the 9th Python in Science Conference*, pp. 56–61. DOI: 10.25080/Majora-92bf1922-00a.

Pedersen, T. L. (2025). *patchwork: The Composer of Plots*. R package version 1.3.2. DOI: 10.32614/CRAN.package.patchwork.

R Core Team (2026). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria.

Silverman, B. W. (1986). *Density Estimation for Statistics and Data Analysis*. London: Chapman and Hall. DOI: 10.1201/9781315140919.

Sukumaran, J. and M. T. Holder (2010). "DendroPy: A Python Library for Phylogenetic Computing". In: *Bioinformatics* 26.12, pp. 1569–1571. DOI: 10.1093/bioinformatics/btq228.

Virtanen, P., R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, İ. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, and P. van Mulbregt (2020). "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python". In: *Nature Methods* 17.3, pp. 261–272. DOI: 10.1038/s41592-019-0686-2.

Wickham, H. (2016). *ggplot2: Elegant Graphics for Data Analysis*. New York: Springer-Verlag. ISBN: 978-3-319-24277-4.

Wilke, C. O. (2025). *ggridges: Ridgeline Plots in ggplot2*. R package version 0.5.7. DOI: 10.32614/CRAN.package.ggridges.